Explainable AI for Cancer Subtype Prediction from Gene Expression

Literature / Related Work Review | Defense I · October 2026
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Project focus. This review supports the development of a research-use software application for breast cancer molecular subtype prediction from processed RNA-seq-derived gene-expression profiles. GSE96058 is the selected primary dataset. Other cancer types and prognosis prediction are outside the committed scope.

1. Introduction and project objective

Breast cancer is a heterogeneous disease whose tumors differ in gene-expression patterns and molecular characteristics. Foundational studies established expression-defined intrinsic subtypes, while PAM50 provided a reproducible framework for assigning subtype labels from selected gene expression measurements [1–3]. Subsequent research has explored gene-panel reduction, alternative machine-learning algorithms, and the robustness of subtype assignments [6–8,11,15].

Our graduation project will develop and evaluate a trained breast cancer subtype classifier integrated into a Streamlit application. It will accept compatible processed expression profiles, report predictions and class scores, explain individual predictions using SHAP, and present traceable literature-supported information about influential genes. The contribution is a carefully evaluated and usable system, not a new molecular subtype, an original clinical diagnostic test, or a claim that explainable AI itself is novel.

2. Biological foundation and prediction target

2.1 Breast cancer intrinsic subtypes

The commonly discussed intrinsic groups are Luminal A, Luminal B, HER2-enriched, and Basal-like; PAM50 implementations may also assign Normal-like [1–3]. Intrinsic subtypes must not be equated directly with immunohistochemistry (IHC) surrogate categories: for example, Basal-like is not synonymous with triple-negative disease, and HER2-enriched does not always mean clinically HER2-positive. These distinctions matter when comparing studies with different outcome labels.

Our initial experiments will evaluate all five PAM50 labels available in GSE96058. Normal-like will receive separate assessment because its biological interpretation and stability remain debated; it will not be removed simply to improve model scores [11,15].

2.2 PAM50 as a reference-labeling procedure

Parker et al. defined a centroid-based molecular-subtype predictor using a 50-gene signature [3]. A new classifier trained to predict existing PAM50 annotations measures concordance with those reference labels, not independent proof of diagnostic superiority. Cascianelli et al. highlighted the dependence of RNA-seq-based subtyping on expression processing and reference construction [6]. Veerla et al. investigated the stability of PAM50 assignments [15], and Raj-Kumar et al. presented the PCAPAM50 implementation and its comparison with IHC-derived groupings [11]. Together, these studies justify careful label-provenance documentation and special attention to ambiguous subtypes.

3. Gene-expression datasets and current preparation status

3.1 GSE96058 / SCAN-B: selected primary dataset

The NCBI GEO record for GSE96058 describes a population-based breast cancer RNA-seq series with 3,273 primary tumors and 136 technical replicate measurements [5]. The project inspection notebook reports 3,409 measurement columns; removing 136 replicate-suffixed measurements leaves 3,273 primary samples, each matched to one PAM50 label, and 30,865 measured gene features. These are recorded notebook outputs, not independently reproduced performance results.

PAM50 label	Samples
Luminal A	1,657
Luminal B	729
Basal-like	339
HER2-enriched	327
Normal-like	221
Total	3,273

The executed notebook reports expression values as log2(FPKM + 0.1). All 50 PAM50 signature genes are represented after resolving the historical gene symbol ORC6L to ORC6: 49 direct symbol matches and one alias match. Gene-symbol resolution will be documented in the model input schema; the prepared expression matrix and processing requirements will be verified before model training.

Because the class distribution is unbalanced (Luminal A accounts for approximately half of the samples), class-specific evaluation is necessary. Technical replicates and any samples attributable to the same patient must not cross development and test partitions.

3.2 Additional resource for potential external evaluation

TCGA-BRCA provides independently collected breast cancer genomic and molecular information [4]. It may support additional generalization testing if subtype definitions, sample identities, gene identifiers, expression measurement scales and processing pipelines can be adequately harmonized. External validation is a possible extension of evaluation, not an outcome already demonstrated or guaranteed by the present data.

4. Machine learning and gene feature selection

Gene-expression classification typically involves many more features than patient samples, increasing overfitting risk. Two feature-space approaches are therefore proposed: (i) the available PAM50 signature genes as a biologically established benchmark feature panel; and (ii) data-driven subsets selected only within training folds using approaches such as univariate statistical filtering and regularization. Candidate gene counts will be compared experimentally rather than fixed in advance.

Cascianelli et al. found regularized multiclass logistic regression competitive for RNA-seq-derived intrinsic subtype classification [6]. Okimoto et al. investigated reduced PAM50 gene panels, including a 36-gene configuration [7]. Park and Rhee combined RNA-seq and ATAC-seq for signature discovery, using feature ranking and SHAP in a four-subtype task [8]. Their multi-omics discovery steps cannot be reproduced directly using only GSE96058 expression data, but their evaluation logic motivates transparent gene selection.

We will compare regularized logistic regression, support vector machine, random forest and XGBoost on consistent development partitions. Model selection will prioritize cross-validated macro-F1, supported by class-specific precision, recall, F1, confusion matrices, and accuracy. Where score displays are presented as

probabilities, their calibration and interpretation will be assessed. A protected held-out test partition will be used only after model and configuration selection.

To avoid data leakage, patient-level partitioning will precede supervised development; technical replicates will be excluded or grouped; and learned scaling, feature selection, filtering and any class-balancing operations will be fitted within the relevant training fold. Using the PAM50 genes as a feature panel must not be described as reproducing Parker et al.'s original centroid classifier [3].

5. Deep learning in related work

Deep learning provides alternative representations of high-dimensional expression data, but it is not an obligatory component of this project. Rostami et al. investigated an autoencoder-related strategy for breast cancer classification, with target groups not directly equivalent to PAM50 intrinsic classes [10]. Shibahara et al. described a neural approach yielding patient-specific interpretable linear models for molecular-subtype prediction [9]. These studies demonstrate possible alternatives while emphasizing that model complexity does not guarantee superior performance or clearer gene-level explanations. Our committed experiments focus on the four classifiers specified above.

6. Explainable AI and biological interpretation

6.1 SHAP attribution

SHAP allocates contributions to input features relative to an explanation baseline and a defined model output [12]. For an individual breast cancer sample, the application will show which measured genes move the predicted-subtype model score upward or downward; attribution units must match the chosen explainer and model output. Global rankings across evaluation samples will be presented separately from case-specific explanations. LIME is a relevant alternative in the literature, but is not the primary method selected for implementation [13].

6.2 Explanation validation and scientific evidence

Explanations that appear plausible are not necessarily reliable. Saliency-map sanity checks by Adebayo et al. provide a methodological warning, although their findings do not directly establish failure of SHAP [14]. We will check whether the explanation reconstructs the correct model output and examine the stability of influential-gene rankings under sensible training variations. Correlated gene-expression features and background-reference choices can affect attribution.

The application will distinguish the predicted class, the gene contributions to the trained model, and published biological associations. Curated evidence will include gene identity, reported function or breast cancer relevance, verified citation and important limitations. The system must state when supporting evidence is unavailable. Neither SHAP attribution nor a published association, by itself, establishes biological causation in an individual sample.

7. Comparative synthesis of relevant studies

Study	Main contribution	Implication for this project
Parker et al. [3]	PAM50 centroid-based intrinsic classification	Reference labeling and PAM50-feature benchmark are different methods
Cascianelli et al. [6]	RNA-seq classifiers and reference construction	Regularized logistic regression baseline; processing compatibility
Okimoto et al. [7]	Reduced PAM50-gene subsets	Compare feature counts without fixing a published subset
Park and Rhee [8]	RNA/ATAC-based feature selection and SHAP	Gene importance precedent; upstream multi-omics not replicated
Shibahara et al. [9]	Interpretable deep-learning architecture	Alternative approach; not required for the prototype
Veerla et al. [15]	PAM50 assignment perturbation and stability	Investigate borderline and Normal-like predictions
Raj-Kumar et al. [11]	PCAPAM50 implementation and label agreement	Reference-procedure sensitivity
Lundberg and Lee [12]	SHAP explanation framework	Class-specific feature attribution and validation

Reported results across publications are not treated as a direct performance ranking: datasets, subtype definitions, preprocessing, and evaluation protocols differ. Our model comparisons will instead use common partitions and outcome definitions within the selected dataset.

8. Research gaps and implications for the proposed system

Existing work establishes the scientific basis for subtype prediction, but integration still requires project-specific evaluation: careful handling of reference PAM50 labels; leakage-safe feature selection; comparisons across several classifiers; performance reporting for all represented subtypes; validated gene-level attributions; and biological statements traceable to publications. A functional Streamlit prototype will connect these elements through single-sample and batch uploads, a consistent prediction pipeline, clear outputs, and downloadable PDF reports.

The initial system is restricted to compatible processed RNA-seq-derived expression profiles rather than raw FASTQ sequencing reads. Uploaded data must match the trained pipeline’s gene identifiers and expression-processing requirements. Results are intended for research and education, not clinical diagnosis or treatment selection.

9. Project limitations and future work

The committed implementation focuses on breast cancer molecular subtype prediction. Broader deployment across RNA-seq studies will require additional input harmonization and independent testing. Possible later extensions include other cancer types and prognosis estimation, which would require new datasets, label definitions and validation protocols. Prognosis is a distinct prediction task requiring appropriate clinical outcomes; subtype scores must not be presented as survival or recurrence-risk estimates.

10. Conclusion

PAM50 and subsequent RNA-seq studies provide a sound basis for evaluating breast cancer subtype classifiers, while the literature highlights dependence on label construction, feature selection and data preparation [3,6–8,11,15]. SHAP can make model behavior more inspectable when explanations are validated and distinguished from biological causation [12–14]. This review therefore supports developing a reproducible breast cancer subtype prediction pipeline and an explainable research-use application with literature-grounded gene interpretation.

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